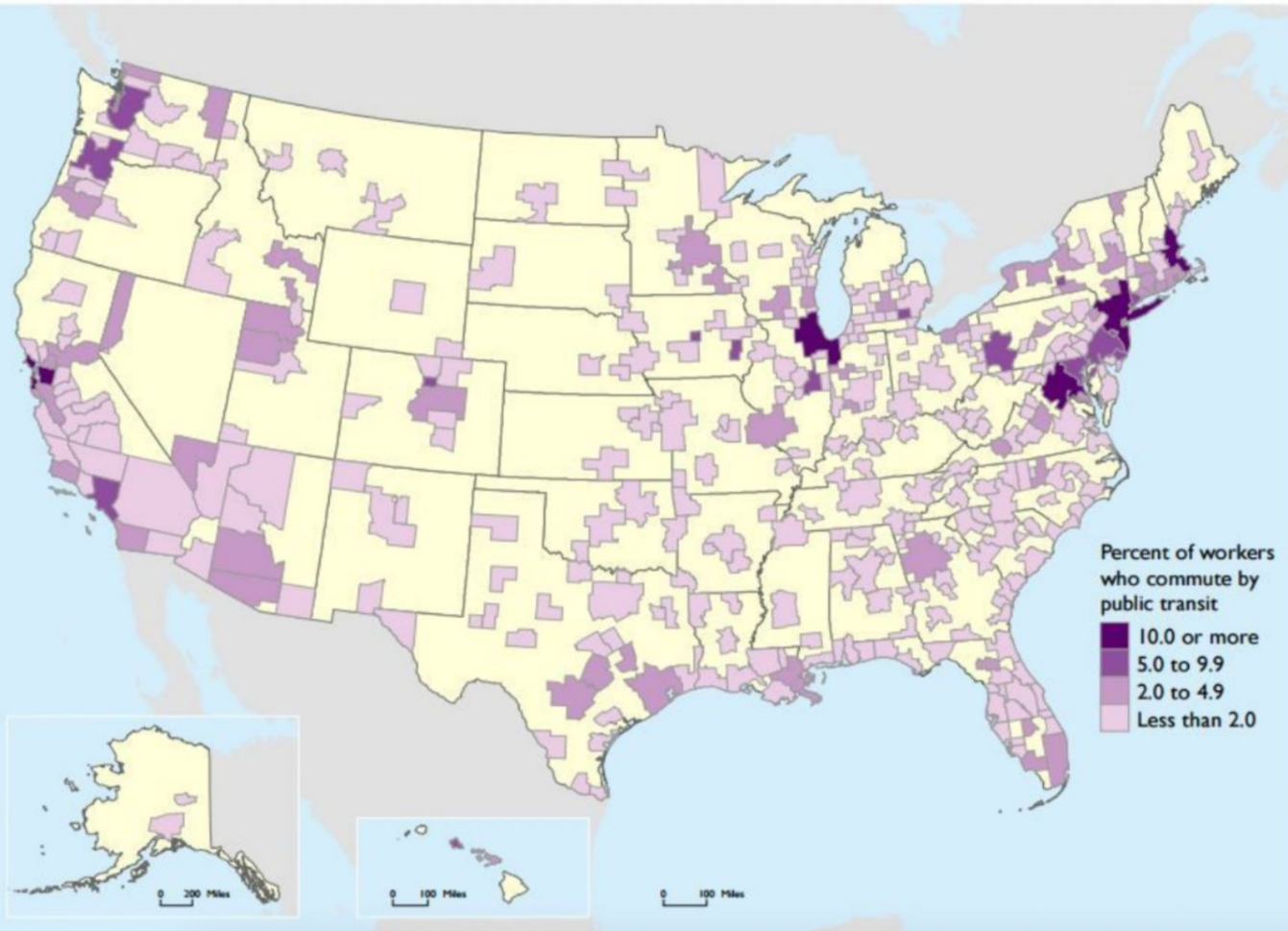


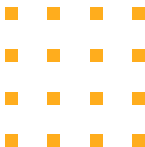
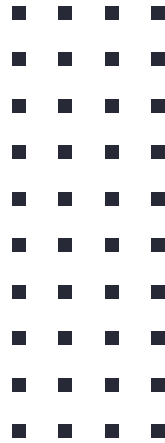
1.1 Introduction to Maps



The map shows the percent of people in metropolitan areas who commuted using public transportation 2010–2014.

A. Identify the map as a *reference* or *thematic* map.

B. Describe spatial patterns in public transportation in American metropolitan areas.



1.4 Spatial Concepts

Objectives and Essential Learning

1.4 Define major geographic concepts that illustrate spatial relationships.

- 1.4.1 Spatial concepts include absolute and relative location, space, place, flows, distance decay, time-space compression, and pattern.



The Spatial Perspective

1. What?
2. Where?
3. Why There?
4. Why Care?
5. Patterns

AP Human Geography Name: _____ Bell _____ pg# _____

 **Thinking Like a Geographer**

Type of Source ☐ map ☐ graph ☐ data ☐ document

Step I: What & Where?	Step II: Patterns
what - where - when - scale - purpose	patterns - similarities - differences - grouping
Step III: Why There?	Step IV: Why Care?
why is that pattern there? - how did it get there? - why does the pattern exist?	impacts - implications - ESPN

What = Place

How do we describe what a location is like?

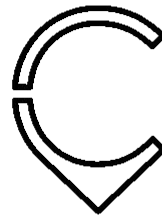
1. Place
 - a. Sense of Place: Factors that contribute to the uniqueness of a location.
 - b. Cultural Landscape/Built Environment: Physical artifacts that humans created which make up the landscape. Human produced.
 - c. Placelessness: A location without a sense of place. No distinct attributes.

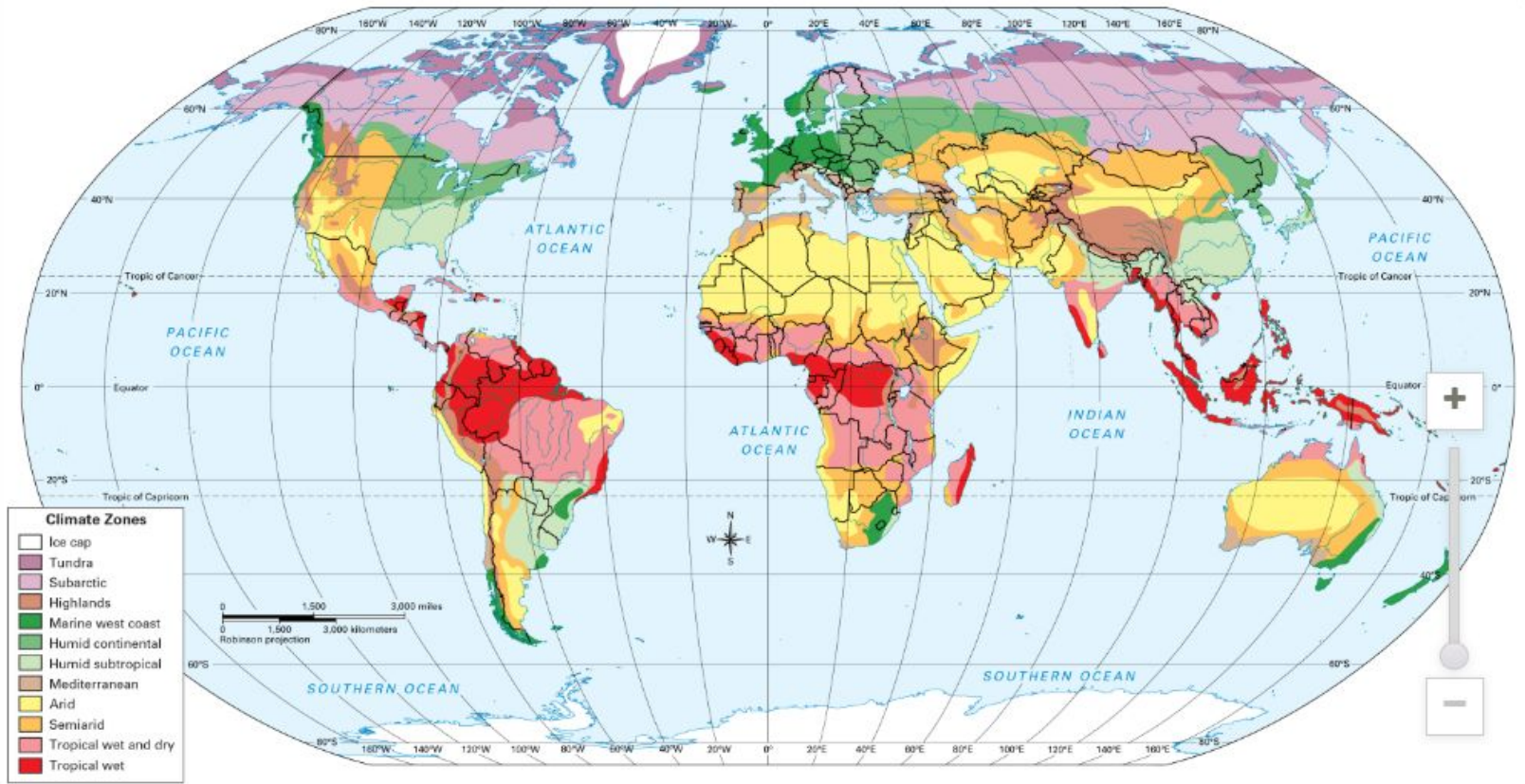


Where = Location

How do we describe where a place is located?

1. Location
 - a. Absolute v. Relative
 - b. Toponym -> Location's name - usually reflective of the culture & history of a place. Ex: Georgetown, Washington, etc.
 - c. Site/Physical Landscape: Environmental features of a location; includes climate, water sources, topography, soil, vegetation, elevation.





Places in tropical latitudes, near the equator, get the most direct rays from the sun all year, so these places have hot weather year-round. Places at high latitudes, close to the North and South poles, receive much less sunlight and remain quite cold all year. Elevation, or altitude, also affects climate. Places at high elevations have colder climates than those lower down.

Patterns – Spatial Analysis

What patterns can be determined?

1. Clustering

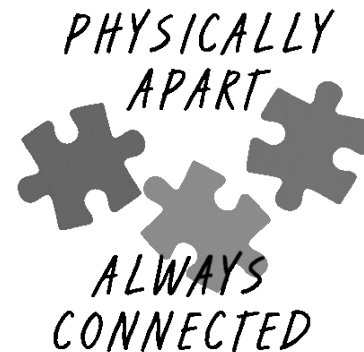
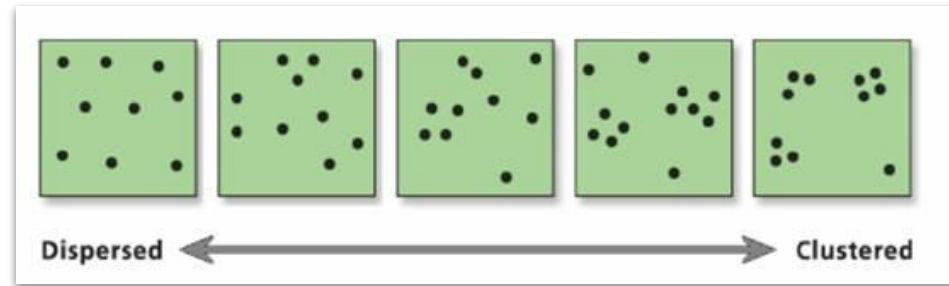
- Close together
- Density - The # of something in a defined area.

2. Dispersal/Distribution

- Far apart
- Distribution - The way something is spread out over an area.

3. Patterns & Spatial Associations

- Indication that two (or more) phenomena may be related, associated, or correlated with one another. RELATIONSHIPS.

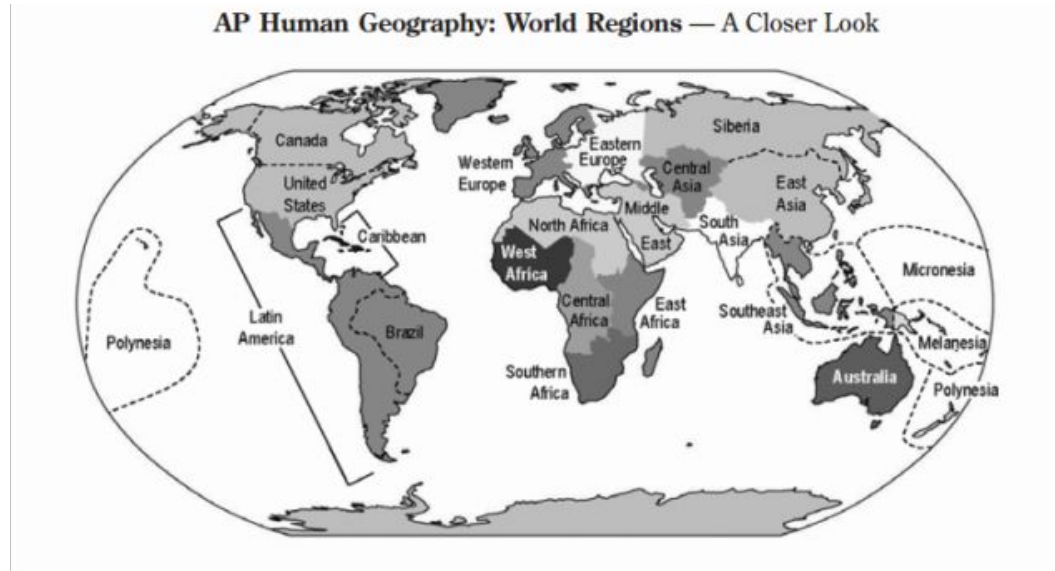


Patterns – Spatial Analysis

What patterns can be determined?

4. Regionalization

- The process geographers use to divide and categorize space into smaller areas of analysis.



Why there? Why care? – Spatial Interactions

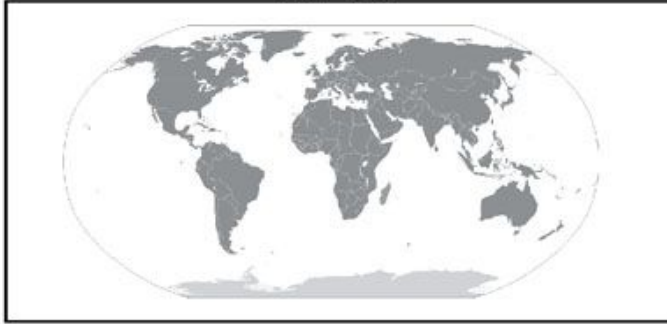
1. Spatial Interaction: Connections, contacts, movement, and flow of things between places.

What geographic concepts do we use to explain the connectedness (or lack thereof) of places, locations, phenomenon?

a. Distance Decay	b. Time-Space Compression
• The interaction between two places declines as the distance between the two places increases. • Physical Barriers-> Mtns./Rivers/Oceans/Deserts • Cultural Barriers -> Language/Religion	• The increasing sense of accessibility and connectivity which seems to bring humans in distance places closer together. • Globalization • Increased technology & transportation



1500–1850



Best average speed of horse-drawn coaches or sailing ships was 10 mph

1850–1930



Steam locomotives averaged 65 mph
Steam ships averaged 38 mph

1950s



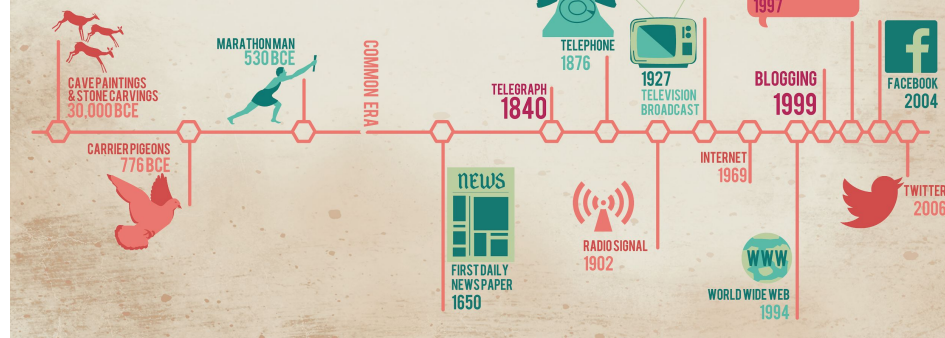
Propellered aircraft 300–400 mph

1960s



Jet passenger aircraft 600–700 mph

THE EVOLUTION OF COMMUNICATION



c. Diffusion: the process by which a characteristic spreads across space from one place to another.

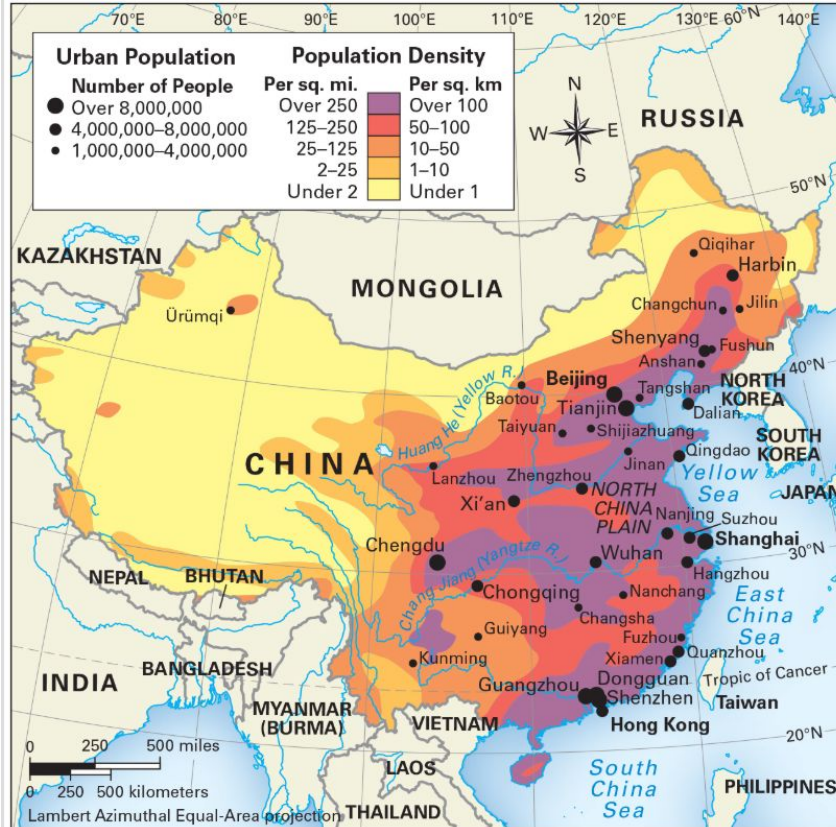
i. Expansion Diffusion

ii. Relocation Diffusion



What? Where? Patterns? Why There? Why Care?

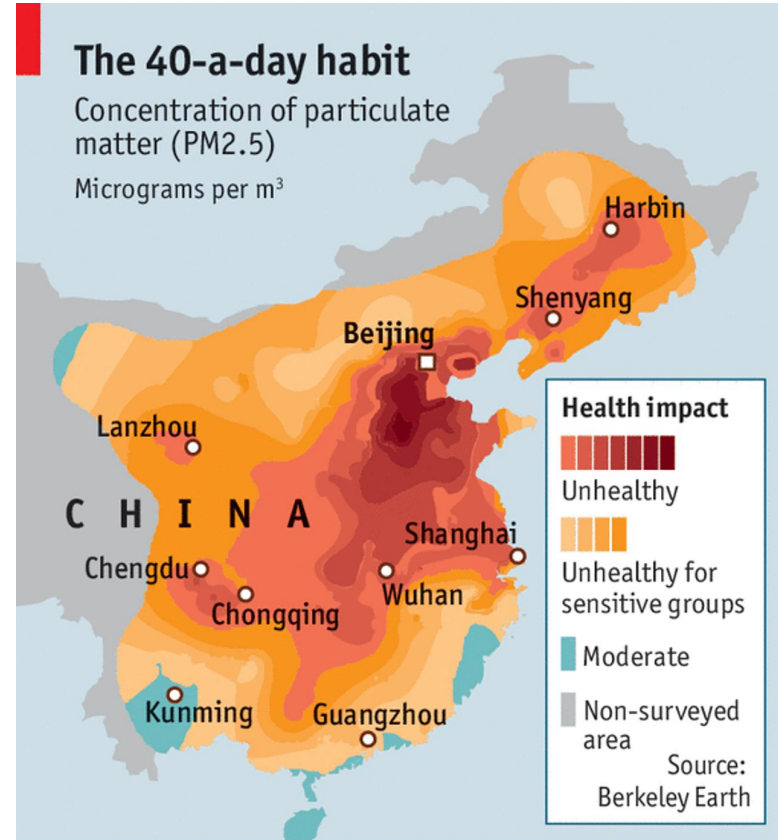
Population Density of China



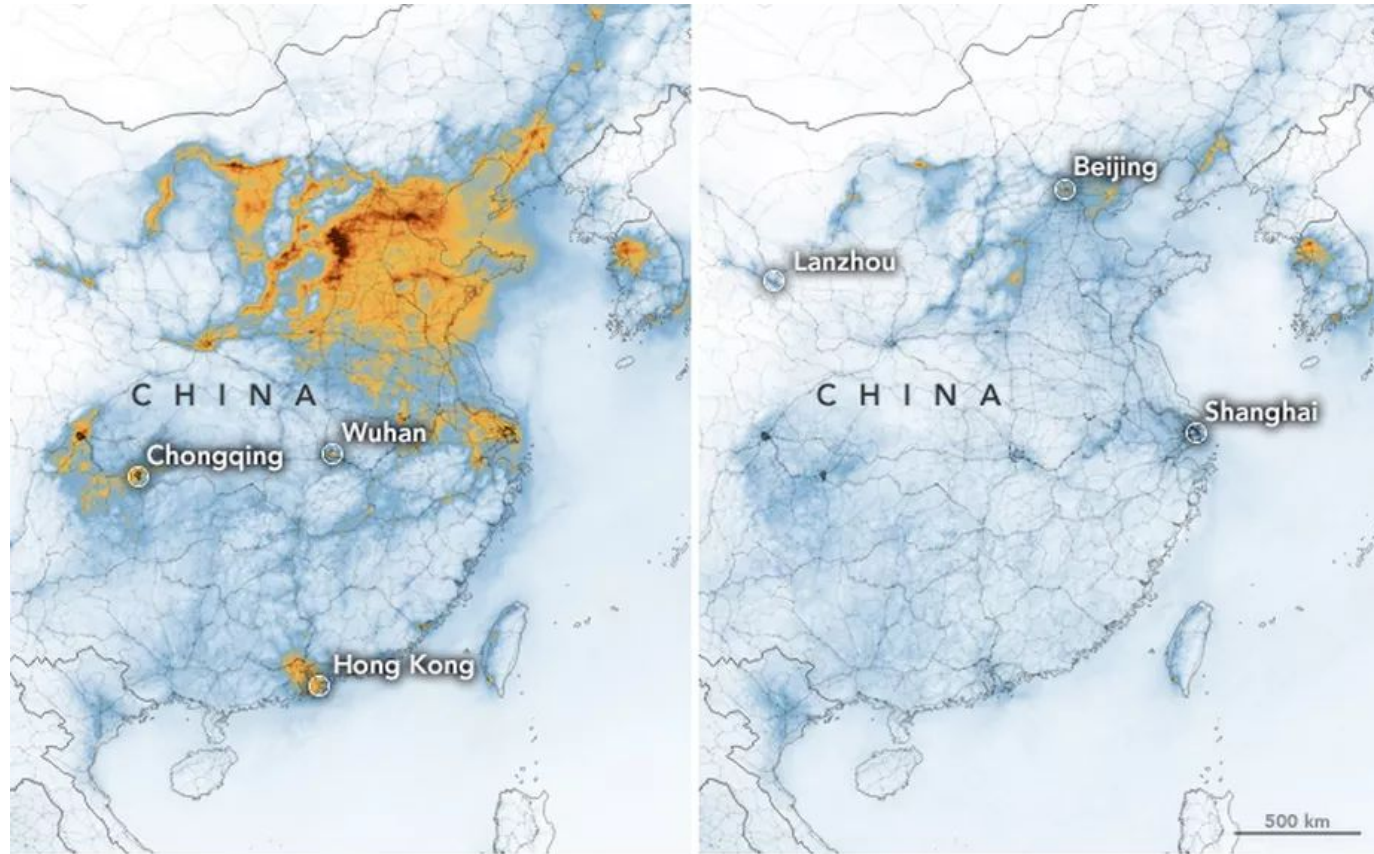
The 40-a-day habit

Concentration of particulate matter (PM_{2.5})

Micrograms per m³



Maps show drastic drop in China's air pollution after coronavirus quarantine



Mean Tropospheric NO₂ Density ($\mu\text{mol}/\text{m}^2$)

Objectives and Essential Learning

1.4 Define major geographic concepts that illustrate spatial relationships.

